Year 9

Mini Test 1

Number and Algebra

Total Marks: 32 marks

Name: ______________________

Total Time: 30 minutes

Show working and answers on this sheet. Show working in sufficient detail to support your answers. Incorrect answers without supporting reasoning may be allocated zero marks.

Questions 1-8 are multiple choice one mark each.
Circle the correct answer.

1. 4^3 in expanded form is:

A 64 **B** $4 \times 4 \times 4$ **C** $3 \times 3 \times 3 \times 3$

D 4^3 **E** 12

2. $2^{12} \div 2^7$ is equal to:

A 4 **B** 8 **C** 16

D 25 **E** 32

3. $2x^7 \times 7x^3$ is equivalent to:

A $14x^{21}$ **B** $14x^{10}$ **C** $14x^4$

D $9x^{10}$ **E** $9x^{21}$

4. $-2x^0 + (9x)^0$ simplifies to:

A −2 **B** −1 **C** 0

D 2 **E** 7

5. $\dfrac{6a^7b^5}{3a^2b^8}$ simplifies to:

A $\dfrac{2a^5}{b^3}$ **B** $\dfrac{b^5}{2a^3}$ **C** $\dfrac{2a^3}{b^5}$

D $\dfrac{2b^3}{a^5}$ **E** $\dfrac{a^5}{2b^3}$

6. When we expand brackets, we use the distributive law. Which of the following does *not* represent the distributive law?

A	$a(b + c) = ab + ac$	**B**	$-a(b + c) = -ab - ac$	**C**	$-a(b - c) = -ab - ac$
D	$-a(b - c) = -ab + ac$	**E**	$a(b - c) = ab - ac$		

7. Which of the following expressions is equivalent to $2(5m + 1)$?

A	$7m + 1$	**B**	$10m + 1$	**C**	$10m + 2$
D	$12m$	**E**	$11m$		

8. Which of the following expressions is equivalent to $5c + 10$?

A	$2c \times 3c + 10$	**B**	$5c + c + 10$	**C**	$15c$
D	$c + 16 + 4c - 6$	**E**	$2c + 16 - 3c - 6$		

9. Evaluate

a. 3^3

b. 7^0

c. 5^{-2}

d. $(2y^0)^3$

[4 marks]

10. Simplify the following expressions

a. $6 \times m \times m \times m$

b. $3bc \times 2ab$

c. $2p - 3p^2 + 7p + 5p^2 - 3$

d. $t^3 \times t^5$

e. $(n^3)^4$

f. $3a^4b^5 \times 7a^7b$

g. $\dfrac{3(e^4)^5 \times e^2}{9e^{-3}}$

[8 marks]

11. Which two algebraic expressions are equivalent?(Circle your choices)

A $7x + 7$ **B** $7x + 7y$ **C** $7xy$ **D** $x + 7y$ **E** $7(x + y)$

[1 mark]

12. Expand the following and simplify when possible.

a. $5(x - 1)$

b. $2x(5 - x)$

c. $10x - 3(2x + 1)$

d. $5(x - 1) + 2(3x - 7)$

[6 marks]

13. Fully factorise the following expressions

a. $6y + 21$

b. $5x^2 - 15x$

c. $-4p^2 - 2p$

d. $3a^2b - 9a^3b^2$

[5 marks]

End of test